

Neil Malan

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EDUCATION

UNIVERSITY OF CAPE TOWN

BSc (HONS, 1ST CLASS), MARINE BIOLOGY AND OCEAN AND ATMOS. SCIENCE

Dec 2011 | Cape Town, South Africa

MSc, OCEAN AND CLIMATE DYNAMICS

Jul 2013 | Cape Town, South Africa

PHD, PHYSICAL OCEANOGRAPHY

Jul 2018 | Cape Town, South Africa

LINKS

Github:// [nmalan](#)

LinkedIn:// [neil-malan](#)

RESEARCH STUDENTS

COMPLETED

Caitlin Sole. **Hons. UCT 2016**

Impact of relative and absolute winds on eddy variability.

Yotam Fadida. **MSc UCT 2019**

Trends in the Agulhas Return Current, published in DSR

IN-PROGRESS

Fernando Sobral. **PhD, UNSW (submitted Dec 2025)**

Cross-shelf heat transport in the East Australian Current

Clea Welch. **MSc, UCT (submitted Dec 2025)**

Subsurface Marine heatwaves in the Mozambique Channel

TEACHING

COURSE CO-CONVENOR

UNSW Sydney 2024-present

- MSCI3001-MSCI5004 - physical oceanography

LECTURER

Rhodes University Dept. Ichthyology 2012-2013

- Teach and coordinate Oceanography module for 2nd and 3rd year Fisheries students.

TEACHING ASSISTANT

University of Cape Town 2014-2017

Sydney Institute of Marine Science 2022

EXPERIENCE

UNIVERSITY OF NEW SOUTH WALES | SENIOR RESEARCH ASSOCIATE, CLIMATE CHANGE RESEARCH CENTRE

July 2023 - Present | Sydney, Australia

- Academic level B position, working with Ass. Prof. Alex Sen Gupta
- Research focus on subsurface marine heatwaves.

UNIVERSITY OF NEW SOUTH WALES | POSTDOCTORAL RESEARCH FELLOW, COASTAL AND REGIONAL OCEANOGRAPHY LABORATORY

Nov 2018 - June 2023 | Sydney, Australia

- Academic level A postdoctoral position in the School of Biological, Earth and Environmental Sciences, working with Professor Moninya Roughan.
- Research focus on cross-shelf Western Boundary Current processes and temperature trends
- Contribute to management of IMOS observation programme as deputy-leader, New South Wales Moorings.

NANSEN-TUTU CENTRE FOR MARINE ENVIRONMENTAL RESEARCH. | POSTDOCTORAL RESEARCH FELLOW

February 2018 - Sep 2018 | Cape Town, South Africa

- Assist in development of base HYCOM model for data operational data assimilation.
- Supervise post-graduate research student.
- Organise international workshop on modelling the Agulhas Current System.

SOUTH AFRICAN ENVIRONMENTAL OBSERVATION NETWORK | DOCTORAL RESEARCH FELLOW - PROFESSIONAL DEVELOPMENT PROGRAM

November 2013 - Sep 2017 | Cape Town, South Africa

- Undertake PhD research.
- Contribute marine expertise to national science meetings.
- Participate in educational programmes for underprivileged schools.

AWARDS

- 2023 Invited Speaker
Gordon Research Seminar on Coastal Ocean Dynamics
- 2021 Best Postdoctoral Research Paper
UNSW CMSI
- 2018 Invited participant, Physical Oceanography Dissertation Symposium
National Science Foundation
- 2016 Best Student Presentation, Agulhas Current Research Symposium
South African Environmental Observation Network
- 2015 SPACES travel fellowship (2-month funded research visit)
GEOMAR centre for ocean research, Kiel.

ACADEMIC SERVICE AND OUTREACH

- **Member**, International Working Group on Marine Heatwaves.
- **Member of Australia's Marine National Facility scientific advisory committee 2024-present**, provide expert advice to the Marine National Facility (MNF) Steering Committee, primarily focused on assessing the research quality of proposed MNF voyages and other matters requiring scientific expertise.
- **Convenor** Oceania Marine Heatwave Seminars. Jan 2025 - present
- **Member of the scientific committee**, 2026 International Conference on Southern Hemisphere Meteorology and Oceanography.
- **Deputy Leader IMOS NSW moorings 2018-2023**, assist in management of mooring program, including contribution to national committees, meeting reporting requirements at both state and national levels, tracking impact and developing data analysis tutorials.
- **Session Convenor or co-convenor** for sessions on Australian Boundary Currents at the Australian Meteorological and Oceanographic Society Annual meetings in 2020 and 2022, as well as Subtropical Western Boundary Current - Shelf interactions at the Ocean Sciences Meeting in 2022.
- **Expert Reviewer** for over 40 articles in journals such as *Geophysical Research Letters*, *Nature Climate Change*, *Journal of Physical Oceanography*, *Global Biogeochemical Cycles* and *Geoscientific Model Development*.

PUBLICATIONS

I have published 20 peer-reviewed journal articles, with 10 as first author and 5 as second author. My articles have received 441 citations with an h-index of 11 (Google Scholar).

- [1] Kathryn E. Smith, Alex Sen Gupta, Michael T. Burrows, Karen Filbee-Dexter, Alistair J. Hobday, Neil J. Holbrook, Neil Malan, Pippa J. Moore, Eric C. J. Oliver, Mads Thomsen, Thomas Wernberg, Zijie Zhao, and Dan A. Smale. "Ocean extremes as a stress test for marine ecosystems and society". In: *Nature Climate Change* (in press). doi: 10.1038/s41558-025-02269-2.
- [2] Neil Malan, Alex Sen Gupta, Amandine Schaeffer, Shujing Zhang, Martina A Doblin, Gabriela Semolini Pilo, Andrew E Kiss, Jason D Everett, Erik Behrens, Antonietta Capotondi, et al. "Lifting the lid on Marine Heatwaves". In: *Progress in Oceanography* (2025), p. 103539.
- [3] Kathryn E Smith, Alex Sen Gupta, Dillon Amaya, Jessica A Benthuyssen, Michael T Burrows, Antonietta Capotondi, Karen Filbee-Dexter, Thomas L Frölicher, Alistair J Hobday, Neil J Holbrook, et al. "Baseline matters: Challenges and implications of different marine heatwave baselines". In: *Progress in Oceanography* 231 (2025), p. 103404.
- [4] Neil Malan, Moninya Roughan, Michael Hemming, and Tim Ingleton. "Quantifying coastal freshwater extremes during unprecedented rainfall using long timeseries multi-platform salinity observations". In: *Nature Communications* 15.1 (2024), p. 424.
- [5] Fernando Sobral, Moninya Roughan, Neil Malan, and Junde Li. "Mean-state and seasonal variability in temperature structure and heat transport in the East Australian Current System from a multi-decadal regional ocean model". In: *Journal of Geophysical Research: Oceans* 129.9 (2024), e2023JC020438.
- [6] Clea Baker Welch, Neil Malan, Daneeja Mawren, Tamaryn Morris, Janet Sprintall, and Juliet Clair Hermes. "Subsurface manifestation of Marine Heatwaves in the South West Indian Ocean". In: *EGU sphere* 2024 (2024), pp. 1–20.
- [7] Michael P Hemming, Moninya Roughan, Neil Malan, and Amandine Schaeffer. "Observed multi-decadal trends in subsurface temperature adjacent to the East Australian Current". In: *Ocean Science* 19.4 (2023), pp. 1145–1162.
- [8] Neil Malan, Moninya Roughan, Michael Hemming, and Amandine Schaeffer. "Mesoscale circulation controls chlorophyll concentrations in the East Australian Current separation zone". In: *Journal of Geophysical Research: Oceans* 128.4 (2023), e2022JC019361.
- [9] Neil Malan, Moninya Roughan, Geoffrey J Stanley, Ryan Holmes, and Junde Li. "Quantifying cross-shelf transport in the East Australian Current system: a budget-based approach". In: *Journal of Physical Oceanography* 52.10 (2022), pp. 2555–2572.
- [10] Lachlan R Phillips, Neil Malan, Moninya Roughan, Robert Harcourt, Ian Jonsen, Martin Cox, Andrew S Brierley, David Slip, Adam Wilkins, and Gemma Carroll. "Coastal seascape variability in the intensifying East Australian Current Southern Extension". In: *Frontiers in Marine Science* (2022).
- [11] Hayden T Schilling, Daniel E Hewitt, Neil Malan, Matthew D Taylor, and Daniel D Johnson. "Cross-jurisdictional larval supply essential for eastern Australian spanner crabs (*Ranina ranina*)". In: *Marine and Freshwater Research* 73.11 (2022), pp. 1352–1367.
- [12] Yotam Fadida, Neil Malan, Meghan F Cronin, and Juliet Hermes. "Trends in the Agulhas Return Current". In: *Deep Sea Research Part I: Oceanographic Research Papers* 175 (2021), p. 103573.
- [13] Neil Malan, Moninya Roughan, and Colette Kerry. "The rate of coastal temperature rise adjacent to a warming western boundary current is nonuniform with latitude". In: *Geophysical Research Letters* 48.3 (2021), e2020GL090751.
- [14] Neil Malan, Matthew Archer, Moninya Roughan, Paulina Cetina-Heredia, Michael Hemming, Carlos Rocha, Amandine Schaeffer, Iain Suthers, and Eduardo Queiroz. "Eddy-driven cross-shelf transport in the East Australian Current separation zone". In: *Journal of Geophysical Research: Oceans* 125.2 (2020), e2019JC015613.

- [15] Jethan Shaun d'Hotman, Neil Malan, Charine Collins, Marc de Vos, Rick Lumpkin, Tamaryn Morris, and Juliet Hermes. "The use of a jet reference frame to analyze drifter trajectories in the Agulhas Current". In: *Journal of Geophysical Research: Oceans* 124.6 (2019), pp. 4238–4247.
- [16] Neil Malan, Jonathan V Durgadoo, Arne Biastoch, Chris Reason, and Juliet Hermes. "Multidecadal wind variability drives temperature shifts on the Agulhas Bank". In: *Journal of Geophysical Research: Oceans* 124.5 (2019), pp. 3021–3035.
- [17] Ahmad Fehmi Dilmahamod, B Aguiar-González, Pierrick Penven, CJC Reason, WPM De Ruijter, N Malan, and JC Hermes. "SIDDIES corridor: a major east-west pathway of long-lived surface and subsurface eddies crossing the subtropical South Indian Ocean". In: *Journal of Geophysical Research: Oceans* 123.8 (2018), pp. 5406–5425.
- [18] N. Malan. "Are icebergs a realistic option for augmenting Cape Town's water supply?" In: *Water Wheel* 17.2 (2018), pp. 32–34.
- [19] Neil Malan, Björn Backeberg, Arne Biastoch, Jonathan V Durgadoo, Annette Samuelsen, Chris Reason, and Juliet Hermes. "Agulhas Current Meanders facilitate shelf-slope exchange on the Eastern Agulhas Bank". In: *Journal of Geophysical Research: Oceans* (2018).
- [20] N Malan, CJC Reason, and BR Loveday. "Variability in tropical cyclone heat potential over the Southwest Indian Ocean". In: *Journal of Geophysical Research: Oceans* 118.12 (2013), pp. 6734–6746.

REFEREES

- Prof. Neil Holbrook (UTAS) neil.holbrook@utas.edu.au
- Prof. Alex Sen Gupta (UNSW) a.sengupta@unsw.edu.au